Artificial and Computational Intelligence
Assignment 2

Problem statement 1 : Gaming

Simulate the working of TicTacToe game extended to three players with coins ‘X’, ‘O’ and ‘T’ with the below sample larger game board of dimension 6*6. The first player to match any four consecutive coins in the same row or same column or same diagonal wins. A sample state of the game board is given below for reference. Your program must start from empty configuration.

X				X	
	X	O		T	
		O	T	T	
		T	T	O	
		X			O

a. You are free to choose your own static evaluation function. Justify your choice of static evaluation value design and explain with a sample game state. Do not use any machine learning model for the evaluation function.

b. Similar to the virtual lab example, one of the players must be a human ie., it must get dynamic inputs from us. The other two players must be simulated using the program.

c. Implement Python code for the design under part a, using Minimax Algorithm.

Problem statement 2: Logic

The cryotherapy data contains details of patients who took this treatment with the following attributes: Sex, Age, Time, Number of warts, Type, Area and the result of treatment (1 successful and 0 not successful). You are required to do the following:

1. Use any of the decision tree algorithms to build a decision tree for the given data.
2. Create rules from the decision tree:
3. Code the rules into a Prolog Knowledge base:
4. Get patient details as input from the user and classify whether the treatment will be successful or not.

Data set :

https://docs.google.com/spreadsheets/d/1fxu2QzydcwipN7i7eqBzVAplU5DIXG0p/edit?usp=sharing&ouid=102856504350644770601&rtpof=true&sd=true

Evaluations will be based on the following.

1. Use Min-Max algorithm and implement the game in PYTHON (35% marks)
2. Derive the rules from the given decision tree and code as Prolog rules. (35% marks)
3. Interactive implementation. Dynamic inputs-based run of the game with step wise board display and error free game ending. (15% marks)
4. Interactive implementation. Dynamic inputs-based run of the logic expert system with step wise options display and error free recommendation & ending. (15% marks)

Important Note:

- You are provided with the python notebook template which stipulates the structure of code and documentation. Use well intended python code.
- Use a separate MS word document for explaining the theory part. Do not include the theory part in the Python notebook except Python comments.
- The implementation code must be completely original and executable.
- Please keep your work (code, documentation) confidential. If your code is found to be plagiarized, you will be penalized severely. Parties involved in the copy will be considered equal partners and will be penalized severely.